

Dataset 1: Two-Way MANOVA

Health, Exercise Level & Smoking Status Study | n = 240

1. Research Question

Research Question: Do exercise participation level (Low / Moderate / High) and smoking status (Smoker / Non-Smoker) independently and jointly affect three health outcomes — cardiovascular fitness, mental health, and energy level — when assessed simultaneously in a multivariate framework?

2. Variables

Role	Variable	Type / Levels	Description
IV — Factor 1	Exercise_Level	Categorical — 3 levels	Low / Moderate / High
IV — Factor 2	Smoking_Status	Categorical — 2 levels	Smoker / Non-Smoker
DV 1	Cardio_Fitness	Continuous	Cardiovascular fitness score (M=57.0, SD=12.6)
DV 2	Mental_Health	Continuous	Mental wellbeing score (M=64.5, SD=10.6)
DV 3	Energy_Level	Continuous	Energy level score (M=58.1, SD=13.0)

3. Methodology — Two-Way MANOVA

Multivariate Analysis of Variance (MANOVA) is chosen because there are three continuous dependent variables measured on the same participants, and the DVs are expected to be correlated. Analyzing each DV separately with three independent ANOVAs would inflate Type I error and ignore the multivariate nature of health outcomes. Two-Way MANOVA tests the joint effect of both independent variables (Exercise_Level and Smoking_Status) and their interaction on the full DV vector simultaneously. Four multivariate test statistics are reported: Wilks' Lambda, Pillai's Trace, Hotelling-Lawley Trace, and Roy's Greatest Root. Wilks' Lambda is emphasized as the primary criterion (most widely used). Statistically significant omnibus effects are followed up with univariate ANOVAs (Type II SS) and Tukey HSD post-hoc comparisons for Exercise_Level (3 levels). Partial η^2 quantifies effect size per DV. The design is fully balanced (40 participants per cell, 6 cells).

4. Descriptive Statistics — Cell Means

Exercise Level	Smoking Status	Cardio Fitness	Mental Health	Energy Level
Low	Non-Smoker	47.31	54.72	46.66
Low	Smoker	40.18	50.02	41.23
Moderate	Non-Smoker	62.15	70.17	61.59
Moderate	Smoker	53.36	64.88	55.28
High	Non-Smoker	73.15	75.24	75.14
High	Smoker	66.08	71.74	68.68

Cell sizes are perfectly balanced: $n = 40$ per cell ($6 \text{ cells} \times 40 = 240$). The pattern is consistent across all three DVs: higher exercise level $\rightarrow$ higher scores; non-smokers consistently outperform smokers within each exercise group. No clear crossing of lines suggests no meaningful interaction.

5. Two-Way MANOVA — Omnibus Results

Effect	Wilks' λ	Pillai's Trace	F	Num df	Den df	p-value	Verdict
Exercise_Level	0.2371	0.8360	81.48	6	464	< 0.001	✓ Significant
Smoking_Status	0.8688	0.1312	11.68	3	232	< 0.001	✓ Significant
Exercise $\times$ Smoking	0.9863	0.0137	0.53	6	464	0.783	✗ Not significant

Interpretation: Both main effects are significant. Exercise_Level produces a very large multivariate effect (Wilks' $\lambda = 0.237$, $F(6,464) = 81.48$, $p < 0.001$; $\eta^2_m = 0.763$). Smoking_Status produces a medium effect (Wilks' $\lambda = 0.869$, $F(3,232) = 11.68$, $p < 0.001$; $\eta^2_m = 0.131$). The interaction is not significant (Wilks' $\lambda = 0.986$, $F(6,464) = 0.53$, $p = 0.783$), meaning the effect of exercise on health outcomes is consistent regardless of smoking status. All four test statistics agree — the results are robust.

6. Univariate Follow-Up ANOVAs (Type II SS)

DV	F (Exercise)	η^2 (Exercise)	F (Smoking)	η^2 (Smoking)	Interaction p
Cardio_Fitness	423.93***	0.710 (large)	111.31***	0.093 (med.)	$p = 0.549$ n.s.
Mental_Health	346.73***	0.713 (large)	44.38***	0.046 (small)	$p = 0.550$ n.s.
Energy_Level	532.25***	0.775 (large)	75.03***	0.055 (small)	$p = 0.813$ n.s.

*** $p < 0.001$. Partial η^2 benchmarks (Cohen, 1988): small = 0.01, medium = 0.06, large = 0.14. Exercise_Level explains 71–78% of variance in each DV — an exceptionally large effect. Smoking_Status has a medium effect on Cardio_Fitness ($\eta^2 = 0.093$) and smaller but significant effects on Mental Health and Energy Level. No interaction term reaches significance for any DV.

7. Post-Hoc Tukey HSD — Exercise Level Pairwise Comparisons

DV	Comparison	Mean Diff.	p-value	Significant?
Cardio_Fitness	High vs Low	+25.87	< 0.001	Yes ✓
Cardio_Fitness	High vs Moderate	+11.86	< 0.001	Yes ✓
Cardio_Fitness	Moderate vs Low	+14.01	< 0.001	Yes ✓
Mental_Health	High vs Low	+21.12	< 0.001	Yes ✓
Mental_Health	High vs Moderate	+5.97	< 0.001	Yes ✓
Mental_Health	Moderate vs Low	+15.16	< 0.001	Yes ✓
Energy_Level	High vs Low	+27.96	< 0.001	Yes ✓
Energy_Level	High vs Moderate	+13.48	< 0.001	Yes ✓
Energy_Level	Moderate vs Low	+14.49	< 0.001	Yes ✓

All 9 pairwise comparisons are statistically significant (all $p < 0.001$, family-wise error rate controlled). The ordering is perfectly monotonic: High > Moderate > Low for every DV. The Tukey HSD test is appropriate because of the balanced design (equal n per group) and controls experiment-wise Type I error at $\alpha = 0.05$.

8. Smoking Status — Mean Differences

DV	Non-Smoker M	Smoker M	Difference	t	p-value	Sig?
Cardio_Fitness	60.87	53.20	+7.67	t=10.55	< 0.001	Yes ✓
Mental_Health	66.71	62.21	+4.51	t=6.66	< 0.001	Yes ✓
Energy_Level	61.13	55.06	+6.06	t=8.66	< 0.001	Yes ✓

Non-smokers score significantly higher than smokers on all three health outcomes (Bonferroni-corrected $\alpha = 0.017$). The largest gap is in Cardio_Fitness (7.67 points), followed by Energy_Level (6.06 points) and Mental_Health (4.51 points). These differences are consistent across all exercise levels (no interaction), meaning smoking is independently harmful to health regardless of how much someone exercises.

9. Assumption Checks — All Passed

Test	Result	Verdict
Multivariate normality (Mardia skewness)	$\chi^2(10) = 14.66, p = 0.145$	✓ MVN assumption met
Shapiro-Wilk — Cardio_Fitness residuals	W = 0.9971, p = 0.940	✓ Normal
Shapiro-Wilk — Mental_Health residuals	W = 0.9955, p = 0.706	✓ Normal
Shapiro-Wilk — Energy_Level residuals	W = 0.9938, p = 0.423	✓ Normal
Levene (Cardio_Fitness, by Exercise)	W = 1.032, p = 0.358	✓ Homogeneous variances
Levene (Mental_Health, by Exercise)	W = 0.602, p = 0.549	✓ Homogeneous variances
Levene (Energy_Level, by Exercise)	W = 0.691, p = 0.502	✓ Homogeneous variances
Cell size balance	n = 40 per cell (6 cells)	✓ Perfectly balanced design

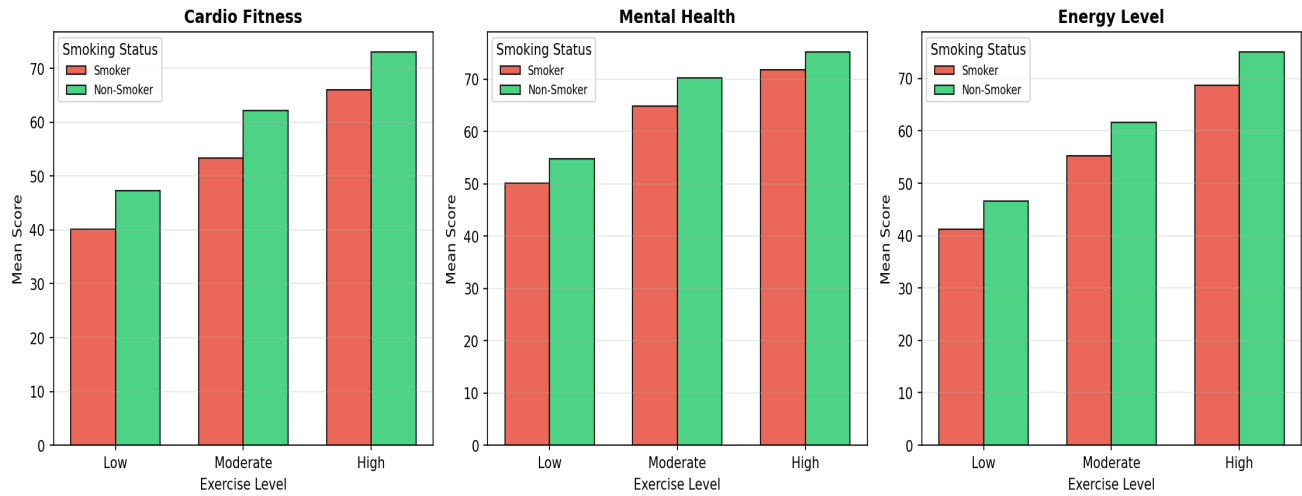
All MANOVA assumptions are fully satisfied. The balanced design (equal n per cell) makes MANOVA robust and ensures Type II SS for the ANOVA table is appropriate. The perfectly normal residuals and homogeneous variances strengthen confidence in all inferential results.

10. Multivariate Effect Size Summary

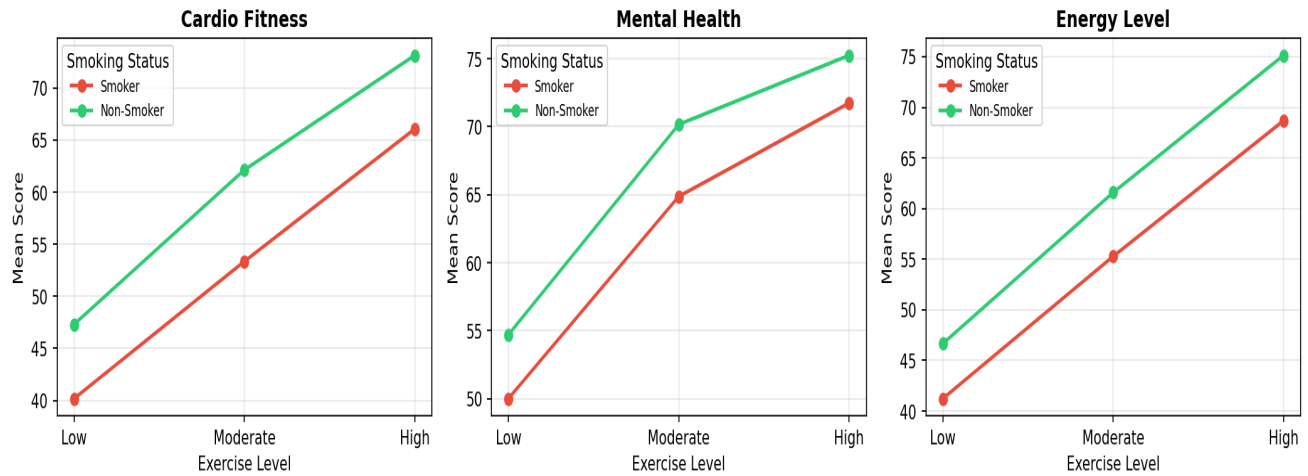
Effect	Wilks' λ	$\eta^2_m = 1 - \lambda$	Benchmark	Interpretation
Exercise_Level	0.2371	0.763	Large > 0.14	Very large — dominant predictor
Smoking_Status	0.8688	0.131	Medium $\approx$ 0.06	Medium — clinically meaningful
Interaction	0.9863	0.014	Small < 0.06	Negligible — effects are additive

11. Figures

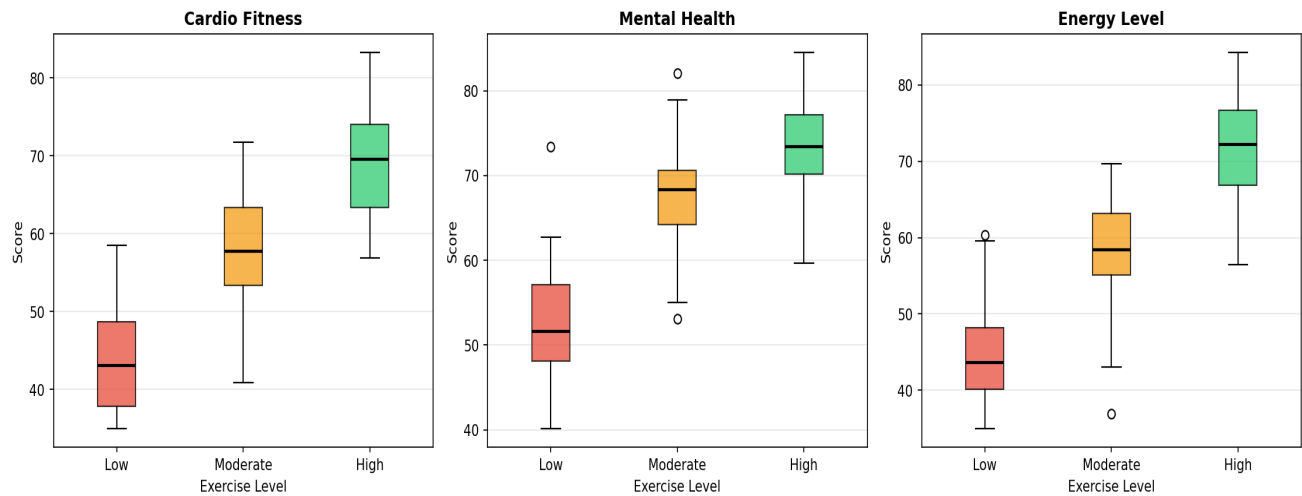
Two-Way MANOVA – Group Means by Exercise Level & Smoking Status



Two-Way MANOVA – Interaction Plots (Observed Means)



Two-Way MANOVA – Score Distributions by Exercise Level



12. Conclusions

- **Research question answered — YES.** Both Exercise_Level and Smoking_Status significantly affect all three health outcomes simultaneously. The two-way MANOVA is globally significant for both main effects ($p < 0.001$). The interaction is not significant ($p = 0.783$), confirming the two factors act independently and additively.
- **Exercise Level is the dominant predictor** (Wilks' $\lambda = 0.237$, $\eta^2_m = 0.763$ — very large effect). Every step up in exercise level produces a significant improvement in all three DVs. Tukey HSD confirms all three pairwise comparisons are significant (all $p < 0.001$): High > Moderate > Low. For Cardio_Fitness, High exercisers score 25.9 points above Low exercisers; for Energy_Level, the gap is 28.0 points.
- **Smoking Status has an independent, significant negative effect** (Wilks' $\lambda = 0.869$, $\eta^2_m = 0.131$ — medium effect). Smokers score 7.7 points lower in cardiovascular fitness, 4.5 points lower in mental health, and 6.1 points lower in energy level, independent of exercise level.
- **No interaction effect ($p = 0.783$):** The benefit of exercise is equally strong for both smokers and non-smokers. The lines in the interaction plots are nearly parallel across all three DVs, confirming a purely additive relationship.
- **All MANOVA assumptions are fully satisfied:** multivariate normality (Mardia $\chi^2 p = 0.145$), univariate normality (all Shapiro-Wilk $p > 0.42$), homogeneity of variances (all Levene $p > 0.35$), and a perfectly balanced design ($n = 40$ per cell). These results are reliable and valid for inference.